

FedLS-SQL: A Novel Federated Large-Small Language Models Framework for Natural Language to SQL

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Abstract. A concise (150-250 words) summary.

I. INTRODUCTION

- **Background**

- Natural Language to SQL (NL-to-SQL) as an interface to relational databases.
- The recent success of LLMs for text-to-SQL.
- The limitations of deploying large models in distributed/resource-constrained environments.
- Privacy concerns when user queries and database schemas cannot be centralized.
- Federated learning (FL) as a solution for decentralized training.

- **Research Gap**

Gap 1: Centralized LLM-based NL-to-SQL

- Strong performance but high computational and memory requirements.
- Requires centralized access to training data.

Gap 2: Federated NL-to-SQL

- Existing FL approaches generally rely on homogeneous client/server models.
- They do not fully exploit the reasoning capabilities of powerful LLMs.

Gap 3: Large-small model collaboration

- Existing teacher-student/distillation approaches are predominantly centralized.
- There is limited investigation of LLM-SLM collaboration within a federated NL-to-SQL setting.

- **Proposed Idea:** A federated large-small language model framework in which a powerful server-side LLM provides SQL reasoning/knowledge guidance, while resource-efficient SLMs perform local privacy-preserving adaptation at clients.

- **Research Questions**

- RQ1: Can large-small language model collaboration improve federated NL-to-SQL accuracy?
- RQ2: Can FedLS-SQL achieve competitive performance with substantially lower client-side computational requirements than large-model FL?
- RQ3: How does server-side LLM guidance affect convergence and generalization under non-IID client data?
- RQ4: What are the trade-offs among SQL accuracy, communication cost, inference latency, and computational resources?

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- **Main Contributions**

- A novel federated large-small language model architecture for NL-to-SQL.
- A large-to-small knowledge transfer mechanism that exploits an LLM without requiring clients to host the LLM.
- A parameter-efficient federated optimization strategy for SLMs.
- Comprehensive experiments covering accuracy, communication efficiency, computational cost, and non-IID robustness.

II. RELATED WORK

2.1 Natural Language to SQL

2.2 Federated Learning for NLP

2.3 Large Language Models for NL-to-SQL

2.4 Large-Small Model Collaboration

III. PROPOSED APPROACH

3.1 Problem Formulation

- Problem Formulation
- Privacy Constraint
- Heterogeneous Model Setting

3.2 Overall Architecture

Four components:

1. Client-side SLM
2. Federated training server
3. Server-side LLM
4. Large-to-small knowledge transfer module

3.3 Client-Side NL-to-SQL Modeling

3.4 Parameter-Efficient Local Adaptation

3.5 Large-to-Small Knowledge Transfer

3.6 Federated Optimization

3.7 Privacy-Preserving Mechanism

4. EXPERIMENTS

4.1 Experimental Setup

- Datasets:

- Spider: Primary cross-domain NL-to-SQL benchmark.
- Spider Realistic: Robustness/generalization evaluation.

- Federated Data Partitioning

- Setting A: IID (Randomly distribute examples)
- Setting B: Domain-based non-IID (Different clients receive different database domains)
- Setting C: Quantity skew (Clients have different numbers of training examples)
- Setting D: Label/SQL-pattern skew (Clients specialize in different SQL structures)
 - JOIN
 - GROUP BY
 - nested queries
 - aggregation
 - filtering

- Server/LLM:

- 7B;
- 8B;
- 14B.

- Client/SLM:

- 0.5B;
- 1.1B;
- 1.5B;
- 3B.

- Model Baselines:

Compare against:

- Centralized SLM
 - SLM trained centrally.
- Federated SLM
 - FedAvg + SLM
 - FedProx + SLM
- Federated PEFT
 - FedAvg + LoRA
- Federated LLM
 - Large model + FL, where computationally feasible.
- LLM Prompting
 - zero-shot LLM
 - few-shot LLM
- Proposed
 - **FedLS-SQL**

- Evaluation Metrics:

- Accuracy Metrics:
 - Exact Match (EM)

- Execution Accuracy (EX)
- Efficiency Metrics:
 - training time;
 - inference latency;
 - GPU memory;
 - CPU memory;
 - number of trainable parameters;
 - communication volume;
 - number of communication rounds;
 - convergence speed;

- Implementation details

4.2 Results

- Overall NL-to-SQL Performance

Method	EM	EX	Trainable Params	Latency
Centralized SLM				
FedAvg-SLM				
FedProx-SLM				
Federated LLM				
LLM Few-shot				
FedLS-SQL				

- Communication Efficiency

- Resource Efficiency

- Non-IID Robustness: Analyze performance under increasingly heterogeneous clients.

- Convergence Analysis: This can demonstrate whether LLM guidance accelerates federated convergence.

4.3 Ablation and Analysis

- Remove LLM Guidance
 - FedLS-SQL→FedAvg-SLM
- Remove Structural Distillation
- Remove Knowledge Distillation
- Different Teacher Sizes

For example:

- 7B;

- 8B;
 - 14B.
- Different Student Sizes
For example:
 - 0.5B;
 - 1.1B;
 - 1.5B;
 - 3B
- LoRA Rank Sensitivity
Evaluate: $r \in \{4, 8, 16, 32\}$
- Number of Clients
For example: $K \in \{5, 10, 20\}$

4.4 Error Analysis

- Schema Linking Errors: The model selects incorrect tables/columns.
- Join Errors: Incorrect join relationships.
- Nested Query Errors: Failures on subqueries.
- Aggregation Errors:
Incorrect:
 - COUNT
 - SUM
 - AVG
 - MAX/MIN
- Federated Distribution Errors: Analyze cases where local client distributions differ significantly.
- LLM-SLM Knowledge Transfer Failures

4.5 Discussion

- Why Large-to-Small Collaboration Works
- Accuracy-Efficiency Trade-off
- Privacy Considerations
- Limitations

5. CONCLUSION

Summarize:

- The problem.
- FedLS-SQL.
- Large-to-small collaboration.
- Federated PEFT.

- Accuracy improvements.
- Efficiency improvements.

Potential future directions:

- differential privacy;
- secure aggregation;
- adaptive teacher selection;
- personalized federated NL-to-SQL;
- continual federated SQL learning;
- multimodal database interfaces;
- agentic LLM-SLM collaboration.
